

Assignment – Part C: Big Data Analysis

COMP SCI 7209 – Big Data Analysis and Project

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Dataset: ER Wait Time Dataset (Creative Commons, 2025)

1. Problem Overview

This modelling task explores whether hospital and patient-related factors can be used to predict emergency department (ED) wait times. The main objective is to identify most prominent features that contribute most to the long delays in care and whether it is possible to estimate the total time before the treatment starts using the simulated dataset. It is possible to make the better staffing rosters, prepare for busier periods and improve the overall experiences of the patients if we study and understand these delays. (Mike Yi, 2025)

Target Variable **total_wait_time_(min)** - a continuous variable representing how long a patient waited from arrival to being seen by a medical professional.

2. Data Preprocessing

Selected Features

Features were selected based on exploratory visualisations and correlation heatmaps. The following were chosen:

- **Numerical:**
nurse-to-patient_ratio, specialist_availability, hour, month
- **Categorical:**
urgency_level, time_of_day, day_of_week, season, region

Preprocessing Steps

- **Datetime parsing:** Extracted hour, month, and day_of_week from visit_date. (GeeksforGeeks, 2025)
- **Encoding:** Used OneHotEncoder (with drop='first') for categorical features.
- **Scaling:** Applied StandardScaler to numerical columns. (Potter, 2025)
- **Train-Test Split:** Dataset split into 80% training and 20% testing sets to ensure fair evaluation.

3. Model Selection

Two models were selected:

Model	Justification
Linear Regression	A simple, interpretable baseline that assumes linear relationships.
Random Forest	A non-linear ensemble model that captures interactions and variable importance.

Table 1: Models

4. Model Training and Refinement

Linear Regression

Trained directly using the processed pipeline. Used as a benchmark model.

Random Forest (Untuned)

Initially trained with:

- `n_estimators = 100`
- `random_state = 42`

Hyperparameter Tuning (Random Forest)

Used GridSearchCV (5-fold cross-validation) to optimise the following:

- `n_estimators`: [50, 100, 150]
- `max_depth`: [None, 10, 20]
- `min_samples_split`: [2, 5, 10]

This grid helped select the best-performing Random Forest model based on validation RMSE. Moreover, after tuning best parameters were as follows:

- `n_estimators`: 150
- `max_depth`: 10
- `min_samples_split`: 10

5. Performance Evaluation

Model	RMSE	R ² Score
Linear Regression	24.75	0.8683
Random Forest (Base)	18.78	0.9242
Tuned Random Forest	18.37	0.9274

Table 2: Model Evaluation

Interpretation

- Both baseline models performed well but Random Forest clearly outperformed as shown in the Table 2.
- The tuned Random Forest model had the lowest RMSE and highest R^2 , suggesting strong predictive power.
- Feature encoding and pipeline design contributed to stable performance.

Visual Evaluation

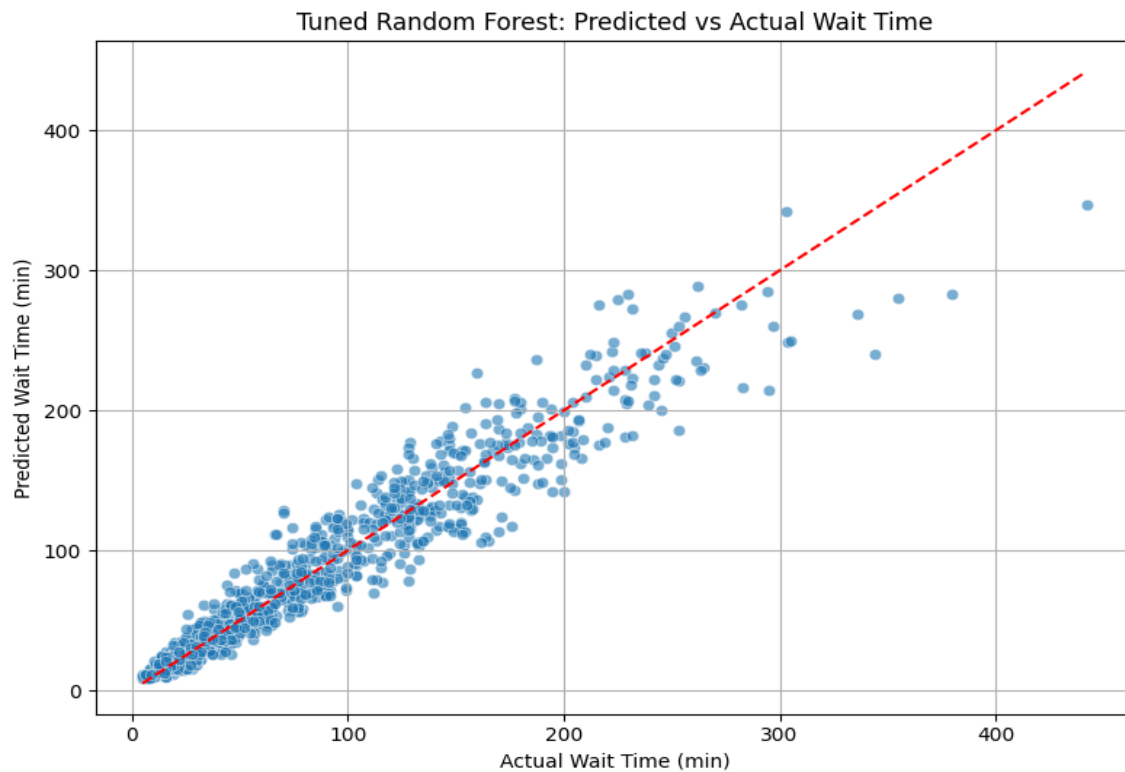


Figure 1: Tuned vs Baseline Random Forest (Mike Yi, 2025)

A scatterplot of predicted vs actual wait times showed alignment along the diagonal for the tuned Random Forest, confirming good accuracy. Minor mispredictions were observed for extremely high wait times, possibly due to synthetic data noise. Moreover, the red line represents the best fit line for the dataset.

6. Model Insights and Future Directions

Observations:

- Urgency level, time of day and day of week strongly influenced waiting time.
- Nurse-to-patient ratio showed a weak but consistent relationship.
- Real-time features like current queue length were missing but would likely improve prediction.

Future Improvements:

- Include features like triage category, arrival method and patient demographics.
- Test additional models like XGBoost or Neural Networks.
- Use cross validation for both models for fair comparison.
- Use real-world dataset from a public hospital system to evaluate the robustness and even train the models using those dataset

7. Conclusion

This stage of the project successfully built and evaluated machine learning models for predicting ED wait times. The tuned Random Forest performed best, achieving an RMSE of 18.37 minutes and an R^2 of 0.9274 on test data. Even though the performance was good, it could be improved using real data. Moreover, the pipeline and modelling workflow are robust and transferable to real-world healthcare analytics projects.

8. References

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GitHub Repository link:

<https://github.com/preheriaa/Big-Data-Analysis-and-Project>